

5,7,12/11/2025

Mechanics

Bionics Eng.

Lectures 10-11

6-8 /x1/2024 

(two lectures 3+2 h)

Constitutive Equations

We will use three types of constitutive equations

1) Rigid rod (inextensible and inflexible) see exercises last week

$$\varepsilon = 0$$
$$\kappa = 0$$

N not determined by const. eqn's, but by equilibrium
 M

$$N = EA \varepsilon$$
$$M = EI \kappa$$

these may stay finite
but are not determined
by ε, κ $\rightarrow +\infty$ $\rightarrow 0$

2) wires (today): cannot sustain compressive forces or bending moments

$$N \geq 0$$
$$M = 0$$
$$\varepsilon = \begin{cases} 0 & \text{if inextensible, then } N \text{ undetermined} \\ N/EA & \text{if extensible} \end{cases}$$

κ not determined

3) Generic linear and uncoupled relations

$$N = EA \varepsilon$$

$$M = EI \kappa$$

see Galileo's and Euler's problem (elasticity). We will return to this later.

Complements ou constitutive questions

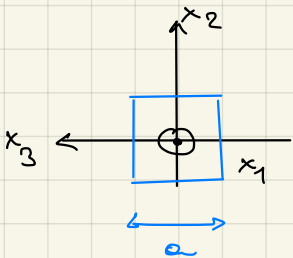
Generally, we will use linear uncoupled equations suitable for symmetric con-
sections

$$\begin{bmatrix} N \\ M \end{bmatrix} = \begin{bmatrix} EA & 0 \\ 0 & EI \end{bmatrix} \begin{bmatrix} \epsilon \\ \kappa \end{bmatrix},$$

$$N = \exists! u' \quad \text{in the small def. theory}$$

where $\bar{t}$ is Young modulus (FL^{-2}), A area of cross section (L^2), I moment of inertia of cross section w/r to bending axis (L^4). For some typical symmetric cross sections:

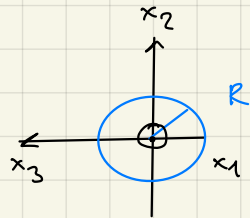
Square



$$A = a^2$$

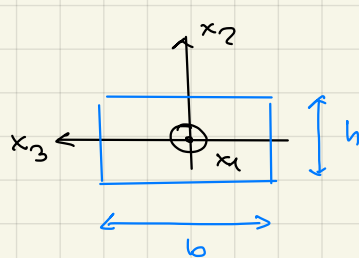
$$I = \int_{-a/2}^{a/2} \int_{-a/2}^{a/2} x_2^2 dx_2 dx_3$$

$$= a \frac{x_2^3}{3} \Big|_{-\frac{2}{12}}^{\frac{2}{12}} = \frac{2^4}{3} \left(\frac{1}{8} + \frac{1}{8} \right) = \frac{2^4}{12}$$

Circle

$$A = \pi R^2, \quad I = \pi \frac{R^4}{4}$$

rectangle

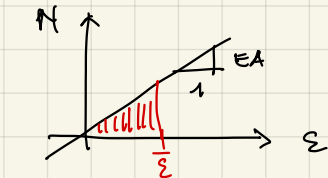


$$A = bh, \quad I = \frac{bh^3}{12}$$

These relations can be derived from 3D elasticity

or by experimental measurements ($N = \overset{\text{material}}{EA} \epsilon$, proportionality N vs ϵ)
↳ geometry of cross-section

These relations can be written in terms of elastic energy densities (energy p.u. length $= \frac{\pi L}{L}$)



$$N(\varepsilon) = EA\varepsilon$$

$$W_{ex}(\varepsilon) = \int_0^\varepsilon N(\eta) d\eta$$

$$= EA \int_0^\varepsilon \eta d\eta = \frac{1}{2} EA \varepsilon^2$$

Then $N(\varepsilon) = W'_{\text{ex}}(\varepsilon)$

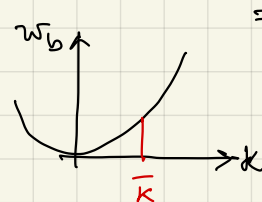


$$M(K) = EI K$$

$$\Rightarrow K \quad W_b(K) = \int_0^K \gamma(y) dy$$

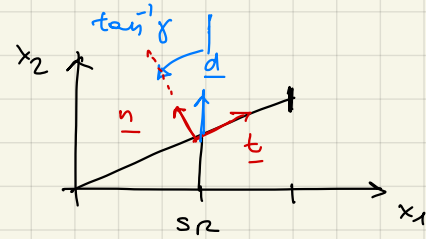
$$= EI \int_0^K \gamma dy = \frac{1}{2} EI K^2$$

Then $M(k) = W'_b(k)$



Ours is a model for "unshearable" rods.

Shearable rods



d director in the plane of the cross-section
 γ shear strain

$\tan^{-1} \gamma$ angle between $\underline{d}$ and $\underline{n}$

For us

$$\underline{d} \equiv \underline{n} \Rightarrow \gamma = 0$$

(the cross-section stays perpendicular to the rod axis even after deformation)

$$\begin{bmatrix} N \\ M \\ T \end{bmatrix} = \begin{bmatrix} EA & 0 & 0 \\ 0 & EI & 0 \\ 0 & 0 & \gamma GA \end{bmatrix} \begin{bmatrix} \epsilon \\ \kappa \\ \gamma \end{bmatrix}$$

G shear modulus (FL^{-2}) (material)

γ shear factor (non-dim) (geometry)

$$\begin{bmatrix} N \\ M \\ T \end{bmatrix} = \begin{bmatrix} EA & 0 \\ 0 & EI \end{bmatrix} \begin{bmatrix} \epsilon \\ \kappa \end{bmatrix} \quad \gamma = 0$$

not determined by const. eq. (reaction to constraint $\gamma = 0$), but by equl. equations. E.g., in linear theory

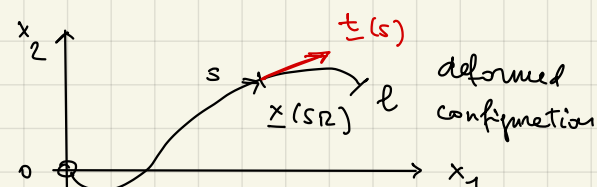
$$T' + f_2 = 0 + BCs \quad \text{or}$$

$$T = -M' - m = -EI w''' - m$$

With spontaneous bending and sketch see later

Cables (or tendons)

Remark: wires as mobile systems: deployable structures
manipulation + intuition + calculation



deformed
configuration

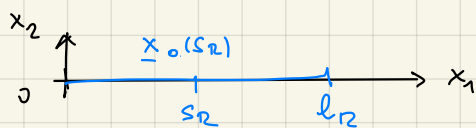
Loads: forces
not couples
(related to the fact that
a wire cannot sustain
bending moments)

Equilibrium

$$M' + T + m = 0 \quad (*)$$

$$\underline{R}(s) = N(s) \underline{t}(s) + T(s) \underline{n}(s)$$

$$\underline{R}' + \underline{f}_R = 0$$



reference
configuration

Constitutive equations

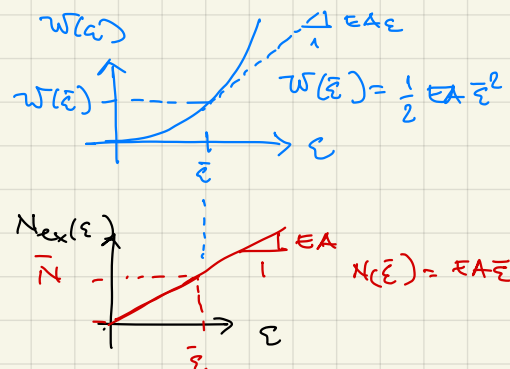
$M \equiv 0$ no resistance to bending $\Rightarrow k$ arbitrary

$N \geq 0$ no resistance to compression

Extensible case: $N = N_{ex}(\epsilon) > 0$ for $\epsilon > 0$
e.g., $EA\epsilon$

Remark: Can find $W(\epsilon)$ s.t. $N_{ex}(\epsilon) = \frac{d}{d\epsilon} W(\epsilon)$. Take $W(\epsilon) = \frac{1}{2} EA \epsilon^2$

Inextensible case: $\epsilon = 0$; $N \geq 0$ undetermined from constitutive equations
and need to use equilibrium equations



$$N = EA \epsilon$$

$\rightarrow$ may stay finite, but not determined by ϵ

Consequences

Let's substitute the const. equ's in our general scheme. $M(s) \equiv 0$, use this in (*)

$$\cancel{M'} + \cancel{T} + \cancel{m} = 0 \Rightarrow T \equiv 0 \Rightarrow \underline{R}(s) = N(s) \underline{t}(s)$$

$$\underline{R}_n(s_R) = N_R(s_R) \underline{t}(s_R)$$

Only N survives as unknown in the internal forces.

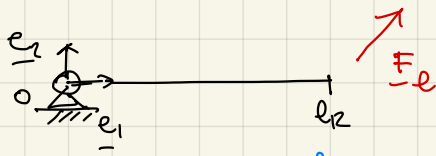
$$\text{From kinematics } \underline{t}(s_R) = \frac{\underline{x}'(s_R)}{|\underline{x}'(s_R)|}$$

$$\text{From equilibrium } \underline{R}' + \underline{f}_R = 0 \Rightarrow \frac{d}{ds_R} (N_R \underline{t}) + \underline{f}_R = 0$$

$$[\underline{N}_R \underline{t}](s_R) + \underline{F}_{s_R} = 0$$

$$\text{BC's } N_R(l_R) \underline{t}(l_R) = \underline{F}_l, \quad N_R(0) \underline{t}(0) = -\underline{F}_0$$

1. Case of no distributed forces in $(0, l_e)$



$$\underline{x}(0) = 0$$

$$\underline{F}_e \text{ given} \Rightarrow \underline{R}_e(l_e) = \underline{N}_e(l_e) \underline{t}(l_e) = \underline{F}_e$$

$$(N_e \underline{t})' = 0 \Rightarrow N_e(l_e) \underline{t}(l_e) - N_e(s_e) \underline{t}(s_e) = 0$$

$$\Rightarrow N_e(s_e) \underline{t}(s_e) = N_e(l_e) \underline{t}(l_e) = \underline{F}_e$$

Case $\underline{F}_e = 0$

$$N_e(l_e) \underline{t}(l_e) = 0 \Rightarrow N_e(l_e) = 0, \underline{t}(l_e) \text{ arbitrary}$$

$$N_e(s_e) \underline{t}(s_e) = 0 \Rightarrow N_e(s_e) = 0, \underline{t}(s_e) \text{ arbitrary}$$

The only geometric restriction comes from $\varepsilon = 0$, true both if inextensible or extensible.

Thus, we have $|\underline{x}'(s_e)| = 1 + \varepsilon(s_e) = 1$

$$|\underline{x}(l_e) - \underline{x}(0)| = \left| \int_0^{l_e} \underline{x}'(\sigma) d\sigma \right| \leq \int_0^{l_e} \underbrace{|\underline{x}'(\sigma)|}_{=1} d\sigma = l_e$$

Remark We also know from $\varepsilon = 0$ that $l = l_e$

Case $\underline{F}_e \neq 0$

$\underline{R}_e: N \geq 0$

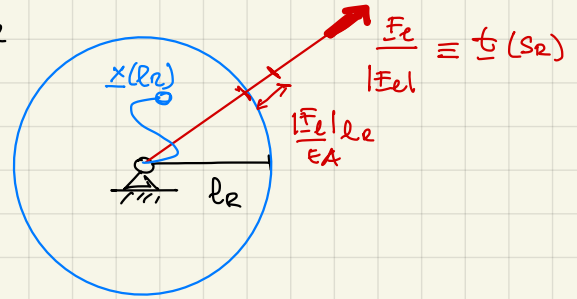
$$N_e(l_e) \underline{t}(l_e) = \underline{F}_e \Rightarrow N_e(l_e) = |\underline{F}_e|, \underline{t}(l_e) = \frac{\underline{F}_e}{|\underline{F}_e|}$$

$$N_e(s_e) \underline{t}(s_e) = \underline{F}_e \Rightarrow N_e(s_e) = |\underline{F}_e|, \underline{t}(s_e) = \frac{\underline{F}_e}{|\underline{F}_e|}, \text{ const.}$$

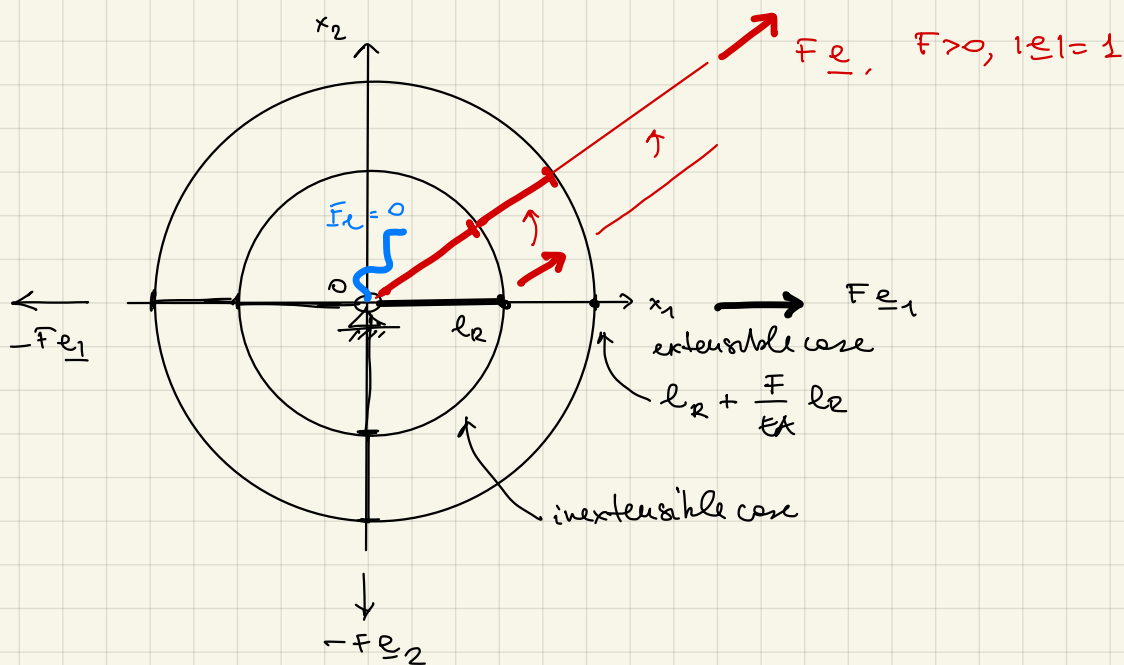
Moreover $\underline{x}'(s_e) = (1 + \varepsilon(s_e)) \underline{t}(s_e) = \underbrace{s'(s_e)}_{=1 + \varepsilon(s_e)} \frac{\underline{F}_e}{|\underline{F}_e|}$

$$\int_0^{s_e} \underline{x}'(\sigma) d\sigma = \underline{x}(s_e) - \underbrace{\underline{x}(0)}_{=0} = \frac{\underline{F}_e}{|\underline{F}_e|} \int_0^{s_e} \underbrace{s'(\sigma)}_{=1 + \varepsilon(\sigma)} d\sigma \Rightarrow \underline{x}(s_e) = 0 + s(s_e) \frac{\underline{F}_e}{|\underline{F}_e|}$$

$$l = \int_0^{l_e} (1 + \varepsilon(\sigma)) d\sigma = \begin{cases} l_e & \text{for } \varepsilon = 0, \text{ inextensible case} \\ \int_0^{l_e} \left(1 + \frac{N_e}{EA}\right) ds_e = l_e + \frac{|\underline{F}_e|}{EA} l_e & \text{for extensible case } \varepsilon = \frac{N_e}{EA} \end{cases}$$

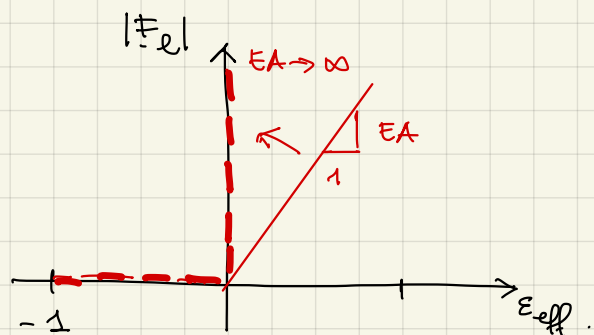


Summary



As a whole structure, it has a response summarized by:

If $|F_e| \neq 0$, then



$$E_{eff} = \frac{d - l_2}{l_2} = \begin{cases} 0 & \text{inert. case} \\ \frac{|F_e|}{EA} & \text{linear, ext. case} \end{cases}$$

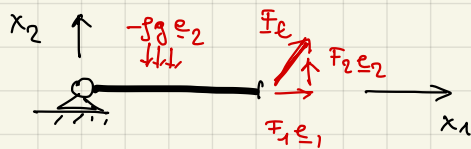
where $d := |x(l_2)|$ is the distance of the free end from the origin (pin-joint).

E_{eff} can be negative if " $d < l_2$ ", which is the behavior we observe when $F_e = 0$.
(distance of $x(l_2)$ from origin $< l_2$)

Remarks

- Control in the region $x(l_2) < l_2$: control l_2
- Muscles and the "push problem": antagonistic muscles

2. Heavy cable (do inextensible case only)



$$\underline{x}(0) = 0, \quad \underline{F}_\ell = F_1 \underline{e}_1 + F_2 \underline{e}_2, \quad \underline{f} = -pg \underline{e}_2$$

$$\varepsilon = 0 \Rightarrow s(s_2) = s_2 \rightarrow \text{drop subscript } \varepsilon \text{ from notation}$$

$$\underline{t}(s) = \underline{x}'(s)$$

$$\text{From } \underline{B}' + \underline{f} = (N\underline{t})' + \underline{f} = 0$$

$$\frac{d}{ds}(N\underline{t}) = pg \underline{e}_2 \xRightarrow{\int ds} \underbrace{N(\ell)\underline{t}(\ell) - N(s)\underline{t}(s)}_{= \underline{F}_\ell} = pg(\ell-s)\underline{e}_2$$

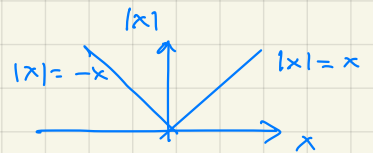
$$\Rightarrow N(s)\underline{t}(s) = F_1 \underline{e}_1 + (F_2 - pg(\ell-s))\underline{e}_2$$

$$\text{Take norm } (N > 0) \quad N(s) = \left[F_1^2 + (F_2 - pg(\ell-s))^2 \right]^{1/2} \quad (*)$$

$$\text{Where } N(s) \neq 0, \quad \underline{t}(s) = \underline{x}'(s) = \frac{1}{N(s)} (F_1 \underline{e}_1 + (F_2 - pg(\ell-s))\underline{e}_2)$$

where $N(s) = 0$, $\underline{t}(s)$ undetermined

$$\text{Case } F_1 = 0 \quad (*) \text{ becomes } N(s) = |F_2 - pg(\ell-s)|$$

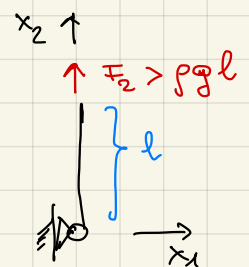


a) $F_1 = 0, F_2 > pg\ell$ (i.e., vertical force stronger than total weight)

$$F_2 - pg(\ell-s) > 0 \quad \forall s$$

$$\Rightarrow N(s) = |F_2 - pg(\ell-s)| = F_2 - pg(\ell-s) > 0 \quad \forall s$$

$$\underline{x}'(s) = \frac{F_2 - pg(\ell-s)}{F_2 - pg(\ell-s)} \underline{e}_2 = \underline{e}_2 \Rightarrow \underline{x}(s) = \underline{x}(0) + s \underline{e}_2$$

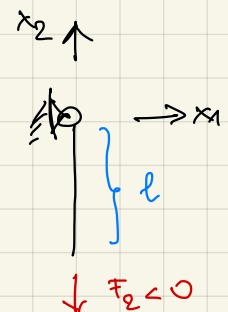


b) $F_1 = 0, F_2 < 0$

$$F_2 - pg(\ell-s) < 0 \quad \forall s$$

$$\Rightarrow N(s) = |F_2 - pg(\ell-s)| = -(F_2 - pg(\ell-s)) > 0 \quad \forall s$$

$$\underline{x}'(s) = \frac{F_2 - pg(\ell-s)}{-(F_2 - pg(\ell-s))} \underline{e}_2 = -\underline{e}_2 \Rightarrow \underline{x}(s) = \underline{x}(0) - s \underline{e}_2$$



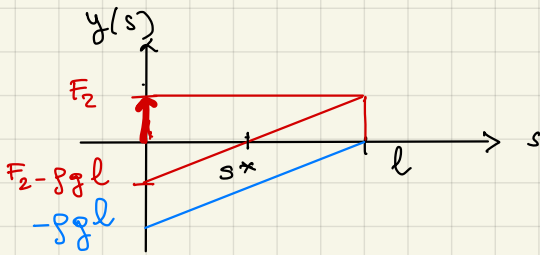
c) $F_1 = 0$, $0 \leq F_2 \leq \rho g l$

In this case, $y(s) := F_2 - \rho g(l-s)$ vanishes/changes sign for some $s^* \in [0, l]$

$$N(s) = |F_2 - \rho g(l-s)| = |y(s)|$$

$$\underline{t}(s) = \frac{F_2 - \rho g(l-s)}{|F_2 - \rho g(l-s)|} \underline{e}_2 \quad (\text{only valid at } s \text{ s.t. } y(s) \neq 0, \text{ i.e., } s \neq s^*)$$

$= \pm 1$ $+1$ where $y(s) > 0$, -1 where $y(s) < 0$.



$$y(s) < 0 \quad \text{for } s < s^*$$

$$y(s) > 0 \quad \text{for } s > s^*$$

$$s^* \text{ s.t. } F_2 - \rho g(l-s^*) = 0$$

$$\Rightarrow F_2 - \rho g l + \rho g s^* = 0$$

$$\Rightarrow s^* = l - \frac{F_2}{\rho g}$$

$$\Rightarrow N(s) = |y(s)| = \begin{cases} -(F_2 - \rho g(l-s)) & s < s^* \\ 0 & s = s^* \\ F_2 - \rho g(l-s) & s > s^* \end{cases}$$

$\underline{x}'(s)$ undetermined for $s = s^*$

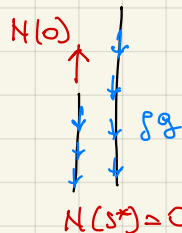
$$\underline{x}'(s) = \frac{F_2 - \rho g(l-s)}{-(F_2 - \rho g(l-s))} \underline{e}_2 = -\underline{e}_2 \quad \text{for } s < s^* \Rightarrow \underline{x}(s) = \underline{x}(0) - s \underline{e}_2, \quad s \in (0, s^*)$$

$$\underline{x}'(s) = \frac{F_2 - \rho g(l-s)}{F_2 - \rho g(l-s)} \underline{e}_2 = \underline{e}_2 \quad \text{for } s > s^* \Rightarrow \underline{x}(s) = \underline{x}(s^*) + (s-s^*) \underline{e}_2 \quad s \in (s^*, l)$$

Free Body Diagram (FBD)

$$\uparrow F_2 < \rho g l$$

$$\uparrow F_2 = N(l)$$



$$N(s^*) = 0$$

$$= \underline{x}(0) - s^* \underline{e}_2 + (s-s^*) \underline{e}_2$$

$$= \underline{x}(0) + (s-2s^*) \underline{e}_2$$

$$= \underline{x}(0) + \left(s - 2l + \frac{2F_2}{\rho g}\right) \underline{e}_2 \quad s \in [s^*, l]$$

Remark Assume $\underline{x}(0) = \underline{0}$, when is $x_2(l) \geq 0$?

$$x_2(l) = \left(-l + \frac{2F_2}{\rho g}\right) \geq 0 \quad \text{if} \quad \frac{2F_2}{\rho g} \geq l$$

$$\text{or } F_2 \geq \frac{1}{2} \rho g l \quad \text{or } s^* = l - \frac{F_2}{\rho g}$$

$$\leq l - \frac{l}{2} = \frac{l}{2}$$

$$N(0) = \rho g l - F_2 = \rho g s^*$$

equilibrium of $(0, s^*)$

$$N(l) = F_2 = \rho g(l-s^*)$$

equilibrium of (s^*, l)

$$+ \rho g(l-s^*) = 0$$

$-\rho g l$
weight

$$\rho g s^*$$

Global equil. $\{ (0, l) \}$

Summary of case c), $F_1 = 0$, $0 \leq F_2 \leq \rho g l$ (upwards force, weaker than total weight)

Keep equ'. $N(s) \underline{t}(s) = \cancel{F_1 e_1} + (F_2 - \rho g(l-s))e_2$ [from integration of $(N\underline{t})' + \underline{f} = 0$]

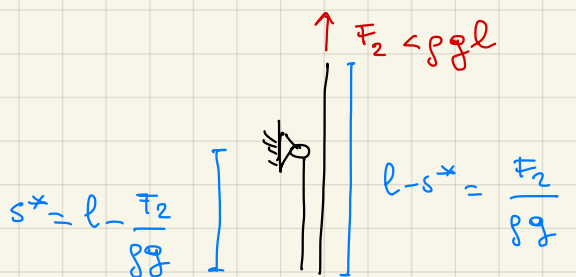
Whenever $N(s) \neq 0$, we can just read-off results from this equation.

$y(s) := F_2 - \rho g(l-s)$ changes sign and vanishes at $s^* = l - \frac{F_2}{\rho g}$

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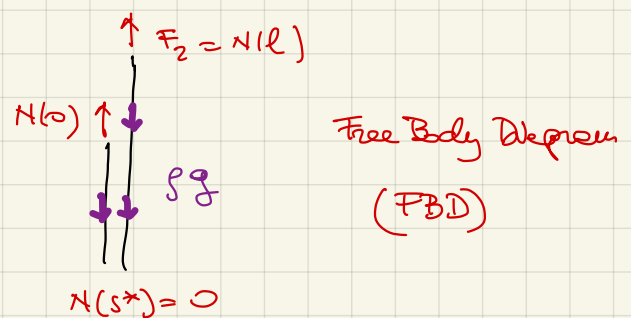
$$N(s) = |F_2 - \rho g(l-s)| = \begin{cases} -(F_2 - \rho g(l-s)) & 0 \leq s \leq s^* \\ 0 & s = s^* \\ F_2 - \rho g(l-s) & s^* \leq s \leq l \end{cases} \quad s^* = l - \frac{F_2}{\rho g}$$

Geometry (equilibrium configuration)



$\underline{t}(s^*)$ undetermined

Statics (Free Body Diagram)



Free Body Diagram
(FBD)

$$N(0) = \rho g s^* \quad N(l) = \rho g(l - s^*)$$

Question when is $x_2(l) \geq 0$?

Question How do you solve a problem in which $x_2(l)$ is given, and $F_1 = 0$?

[To be hired by Google]

Case $F_1 \neq 0$; next lecture

Wires continued

Const. eqn's $\begin{cases} M=0 \\ N \geq 0 \end{cases}$; $m=0 \Rightarrow T \equiv 0$
 $\varepsilon = 0$ (inext. case)

Governing equations for inextensible case ($\varepsilon=0$, $N \geq 0$ unknown)

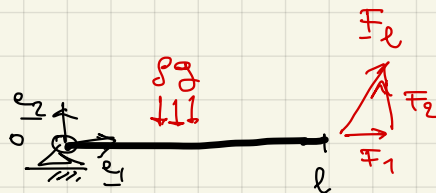
$$\underline{t}(s) = \underline{x}'(s), \quad \underline{R}(s) = N(s) \underline{t}(s)$$

$$(N \underline{t})' + \underline{f} = \underline{0} \quad (*)$$

$$[[N \underline{t}]](s_i) + \underline{F}_{si} = \underline{0}$$

$$N(l) \underline{t}(l) = \underline{F}_l; \quad N(0) \underline{t}(0) = -\underline{F}_0$$

Case of heavy cable $\underline{f} = -\rho g \underline{e}_2$



$$\underline{x}(0) = \underline{0}$$

$\underline{F}_l$ given

$$F_1 \neq 0$$

(eg., $F_1 > 0$)

$\int_s^l ds$ of $(*)$. Get: $N(s) \underline{t}(s) = N(l) \underline{t}(l) + \int_s^l \underline{f}(\sigma) d\sigma = F_1 \underline{e}_1 + (F_2 - \rho g(l-s)) \underline{e}_2$

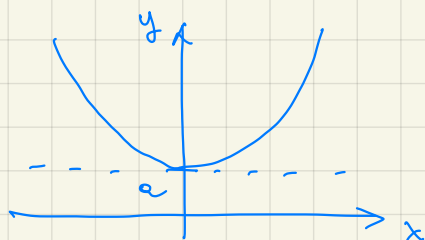
Take norm. Get: $N(s) = \left[F_1^2 + (F_2 - \rho g(l-s))^2 \right]^{1/2} \neq 0 \quad \forall s$

Then $\underline{t}(s) = \underline{x}'(s) = \frac{1}{N(s)} \left[F_1 \underline{e}_1 + (F_2 - \rho g(l-s)) \underline{e}_2 \right]$

$$= \frac{F_1}{\sqrt{F_1^2 + (F_2 - \rho g(l-s))^2}} \underline{e}_1 + \frac{F_2 - \rho g(l-s)}{\sqrt{F_1^2 + (F_2 - \rho g(l-s))^2}} \underline{e}_2$$

Can integrate explicitly and get $x_1(s)$, $x_2(s)$ in terms of hyperbolic trig. functions $\rightarrow$ Catenary

In suitable coordinates, y vertical, x horizontal, $y = a \cosh\left(\frac{x}{a}\right)$, $a = \frac{|F_1|}{\rho g} = \frac{T_0}{\rho g}$ (a has dim. of length)



$$y(x) = a \cosh\left(\frac{x}{a}\right)$$

$$y'(x) = a \frac{1}{a} \sinh\left(\frac{x}{a}\right)$$

$$y''(x) = \frac{1}{a} \cosh\left(\frac{x}{a}\right), \quad y''(0) = \frac{1}{a} = \frac{\rho g}{T_0}$$

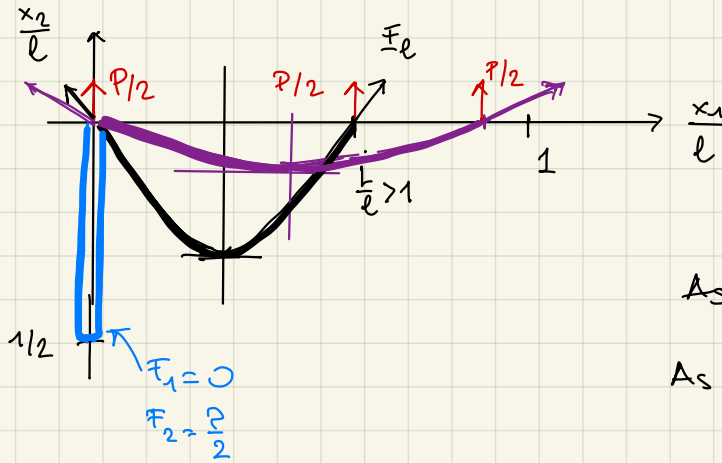
$$T_0 \rightarrow 0 \Rightarrow y''(0) \rightarrow +\infty$$

$$T_0 \rightarrow \infty \Rightarrow y''(0) \rightarrow 0$$

To place this shape in space need BC's.

$$\underline{x}(0) = \underline{0}$$

Consider the case $\underline{x}(l) = L \underline{e}_1, 0 \leq L < l$:



End reactions are now unknown.

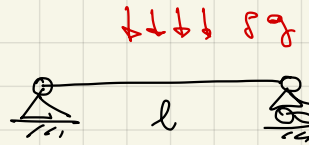
By symmetry, vertical comp's of reactions = $\frac{P}{2}$ each

where $P = pg l$ total weight

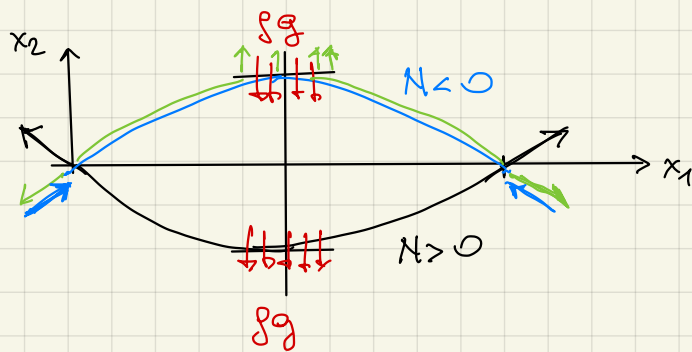
$$\text{As } \frac{L}{l} \rightarrow 1, F_1 \rightarrow \infty$$

$$\text{As } \frac{L}{l} \rightarrow 0, F_1 \rightarrow 0 \quad \text{see case } F_1 = 0, c)$$

Question . What about this case:



Remark Arches under pure compression shaped using the physical model of the catenary
(a form-finding "algorithm")



$$\begin{cases} M = 0 \\ N \geq 0 \end{cases}$$

wire

$$\begin{cases} M = 0 \\ N \leq 0 \end{cases}$$

masonry arch
(no-tension material)

Just change sign to N and the flipped configuration is an equilibrium one for the arch under its own weight, such that $N < 0$, $M \equiv 0$.

Galileo $\rightarrow$ Huygens $\rightarrow$ Hooke
(parabola) (false) (catenary)
 $y \sim x^2$ as a form-finding
analogue
"algorithm"

Ut pendet continuum flexible, sic stabili
continuum rigillum inversum

As a flexible continuum hangs, so stands
a rigid inverted continuum

St. Peter's (Rome) $\rightarrow$ St. Paul's (London) $\rightarrow$ Gaudi's Sagrada Família (Barcelona)
(double sphere) (triple dome, one is $y \sim x^3$) (lots of models in the museum)
When tries to follow Hooke

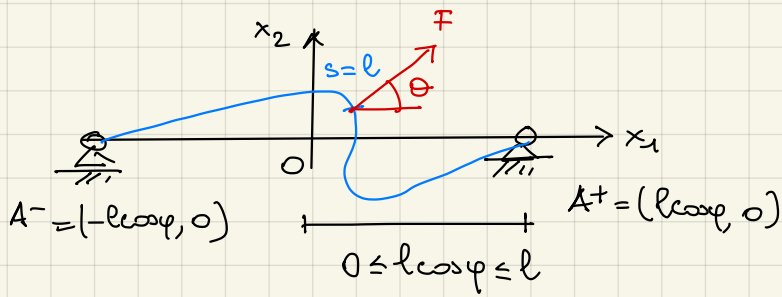
All of this follows from:

$$\underline{t}(s) = \underline{x}'(s), \quad \underline{R}(s) = N(s) \underline{t}(s)$$

$$(N \underline{t})' + \underline{f} = \underline{0} \quad (*)$$

$$N(l) \underline{t}(l) = \underline{F}_e; \quad N(0) \underline{t}(0) = -\underline{F}_0$$

Case of point-force load (no weight, inextensible $\varepsilon = 0$)



length = $2l$, supports at distance $2l \cos \varphi \leq 2l$

$$\underline{x}(0) = -l \cos \varphi \underline{e}_1; \quad \underline{x}(2l) = l \cos \varphi \underline{e}_1, \quad \varphi \in (0, \frac{\pi}{2})$$

$$\underline{F} = F(\cos \theta \underline{e}_1 + \sin \theta \underline{e}_2), \quad F > 0, \text{ applied at } s = l$$

Equations

$$\begin{cases} |\underline{x}'(s)| = 1, & N \geq 0 \\ (N \underline{x}')' = 0 & (*) \\ \llbracket N \underline{x}' \rrbracket(l) = -F(\cos \theta \underline{e}_1 + \sin \theta \underline{e}_2) \\ \underline{x}(0) = -l \cos \varphi \underline{e}_1; \quad \underline{x}(2l) = l \cos \varphi \underline{e}_1 \end{cases}$$

Integrate (*)

taking norms

$$s \in (0, l), \int_0^s ds \rightarrow (N \underline{x}')(s) = (N \underline{x}')(0) = -\underline{F}_0 \Rightarrow N(s) \equiv |\underline{F}_0| =: N^- \text{ (a const.)}$$

$$\text{If } N^- \neq 0, \text{ then } \underline{x}'(s) \equiv -\frac{\underline{F}_0}{N^-} =: \underline{t}^- \text{ (a const.)}$$

$$s \in (l, 2l), \int_s^{2l} ds \rightarrow (N \underline{x}')(2l) = (N \underline{x}')(s) = \underline{F}_{2l} \Rightarrow N(s) \equiv |\underline{F}_{2l}| =: N^+ \text{ (a const.)}$$

$$\text{If } N^+ \neq 0, \text{ then } \underline{x}'(s) \equiv \frac{\underline{F}_{2l}}{N^+} =: \underline{t}^+ \text{ (a const.)}$$

Since $\llbracket N \underline{x}' \rrbracket(l) = N^+ \underline{t}^+ - N^- \underline{t}^- \neq 0$, cannot have $N^+ = N^- = 0$

At least one must be $\neq 0$. Three cases

1. $N^- > 0, N^+ = 0$

2. $N^- = 0, N^+ > 0$

3. $N^- > 0, N^+ > 0$

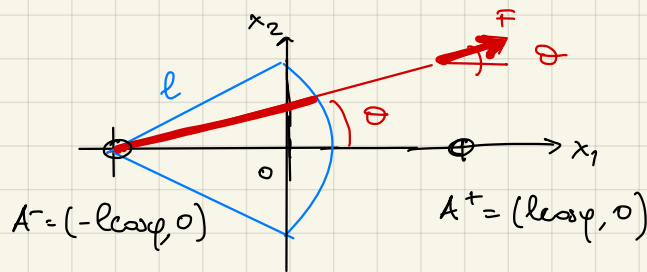
1. $N^- > 0, N^+ = 0$

$$N^+ \underline{t}^+ - N^- \underline{t}^- = -N^- \underline{t}^- = -F(\cos\theta \underline{e}_1 + \sin\theta \underline{e}_2)$$

Take norms $N^- = F > 0$, then $\underline{t}^- = \frac{F}{N^-} (\cos\theta \underline{e}_1 + \sin\theta \underline{e}_2) = \cos\theta \underline{e}_1 + \sin\theta \underline{e}_2$

$$\Rightarrow \alpha(s) \equiv \theta =: \alpha^-, s \in (0, l)$$

Therefore $\underline{x}(l) = \underline{x}(0) + l \cos\theta \underline{e}_1 + l \sin\theta \underline{e}_2$
 $= (l \cos\theta - l \cos\varphi) \underline{e}_1 + l \sin\theta \underline{e}_2$



$N^+ = 0 \Rightarrow \underline{t}^+$ arbitrary.

θ cannot be arbitrary.

From $|\underline{x}(2l) - \underline{x}(l)| = \left| \int_l^{2l} \underline{x}'(s) ds \right| \leq \int_l^{2l} |\underline{x}'(s)| ds = l$

$$l \cos\varphi \underline{e}_1 - l(\cos\theta - \cos\varphi) \underline{e}_1 - l \sin\theta \underline{e}_2 = l[(2\cos\varphi - \cos\theta) \underline{e}_1 - \sin\theta \underline{e}_2]$$

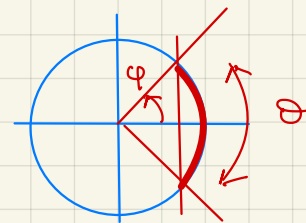
Take norm² $l^2[(2\cos\varphi - \cos\theta)^2 + \sin^2\theta] \leq l^2$

$$\Rightarrow 4\cos^2\varphi - 4\cos\varphi\cos\theta + \cos^2\theta + \sin^2\theta \leq 1$$

$$\Rightarrow 4\cos^2\varphi - 4\cos\varphi\cos\theta \leq 0$$

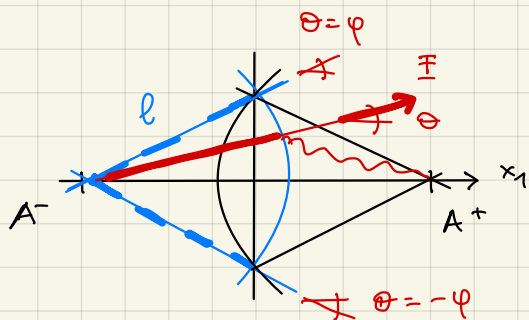
$$\Rightarrow 1 \leq \frac{\cos\theta}{\cos\varphi} \Leftrightarrow \cos\theta \geq \cos\varphi$$

$$\Leftrightarrow \theta \in (-\varphi, \varphi)$$



Get $\theta \in (-\varphi, \varphi)$

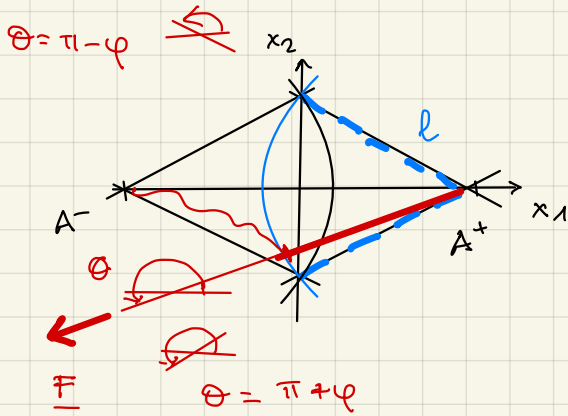
In summary:



There are solutions of this type for

$$\theta \in (-\varphi, \varphi), F > 0 \text{ arbitrary}$$

Case 2, $N^- = 0, N^+ > 0$ is analogous



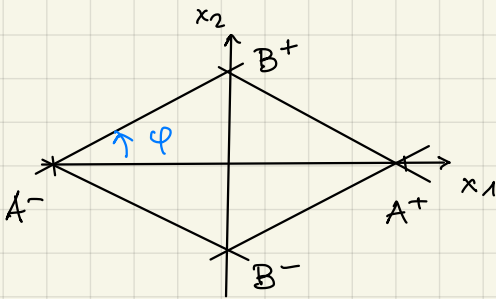
There are solutions of this type for

$$\theta \in (\pi - \varphi, \pi + \varphi), F > 0 \text{ arbitrary}$$

3. $N^+ > 0, N^- > 0$

$$\underline{t}(s) = \begin{cases} \underline{t}^- & s \in (0, l) \\ \underline{t}^+ & s \in (l, 2l) \end{cases}$$

both constant, and the wires of loop in $l \Rightarrow$ two possible soln's
(intersect circles of radius l centered at $A^\pm$)



$$\underline{x}(l) = B^+$$

Case B^+ $\alpha^- = \varphi, \alpha^+ = -\varphi, \underline{x}(l) = l \sin \varphi \underline{e}_2$

Jump cond. where force is applied

$$N^+ (\cos \alpha^+, \sin \alpha^+) - N^- (\cos \alpha^-, \sin \alpha^-) = -F (\cos \theta, \sin \theta)$$

$$\Rightarrow N^+ (\cos \varphi, -\sin \varphi) - N^- (\cos \varphi, \sin \varphi) = -F (\cos \theta, \sin \theta) \text{ and solve for } N^+, N^-, \text{ see } \square \text{ box below}$$

Project $N^+ \cos \varphi - N^- \cos \varphi = -F \cos \theta$

$$-N^+ \sin \varphi - N^- \sin \varphi = -F \sin \theta$$

$$N^+ \cos \varphi \sin \varphi - N^- \cos \varphi \sin \varphi = -F \cos \theta \sin \varphi$$

$$-N^+ \sin \varphi \cos \varphi - N^- \sin \varphi \cos \varphi = -F \sin \theta \cos \varphi$$

$$N^- = \frac{F}{2} \left(\frac{\cos \theta}{\cos \varphi} + \frac{\sin \theta}{\sin \varphi} \right)$$

$$N^+ = \frac{F}{2} \left(\frac{\sin \theta}{\sin \varphi} - \frac{\cos \theta}{\cos \varphi} \right)$$



⊕

$$-2N^- \sin \varphi \cos \varphi = -F (\cos \theta \sin \varphi + \sin \theta \cos \varphi)$$

⊖

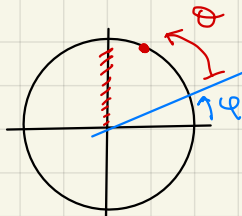
$$2N^+ \sin \varphi \cos \varphi = F (\sin \theta \cos \varphi - \cos \theta \sin \varphi)$$

Restrictions on Θ come from $N^{\pm} > 0$

$$N^{-} = \frac{F}{2} \left(\frac{\cos \Theta}{\cos \varphi} + \frac{\sin \Theta}{\sin \varphi} \right)$$

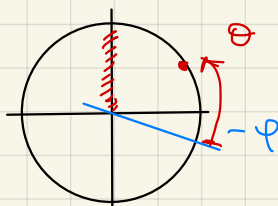
$$N^{+} = \frac{F}{2} \left(\frac{\sin \Theta}{\sin \varphi} - \frac{\cos \Theta}{\cos \varphi} \right)$$

$$\begin{aligned} N^{-} &= \frac{F}{2} \frac{\cos \Theta \sin \varphi + \sin \Theta \cos \varphi}{\cos \varphi \sin \varphi} \\ &= \frac{F}{2} \frac{\sin(\Theta + \varphi)}{\cos \varphi \sin \varphi} \end{aligned}$$



$$\sin(\varphi + \Theta) > 0 \text{ for } \Theta \in (-\varphi, \pi - \varphi)$$

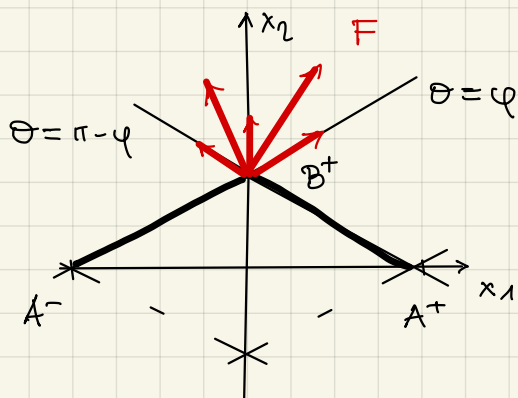
$$\begin{aligned} N^{+} &= \frac{F}{2} \frac{\sin \Theta \cos \varphi - \cos \Theta \sin \varphi}{\sin \varphi \cos \varphi} \\ &= \frac{F}{2} \frac{\sin(\Theta - \varphi)}{\sin \varphi \cos \varphi} \end{aligned}$$



$$\sin(-\varphi + \Theta) > 0 \text{ for } \Theta \in (\varphi, \pi + \varphi)$$

$$\text{Intersecting } (-\varphi, \pi - \varphi) \cap (\varphi, \pi + \varphi) = (\varphi, \pi - \varphi)$$

Summarizing

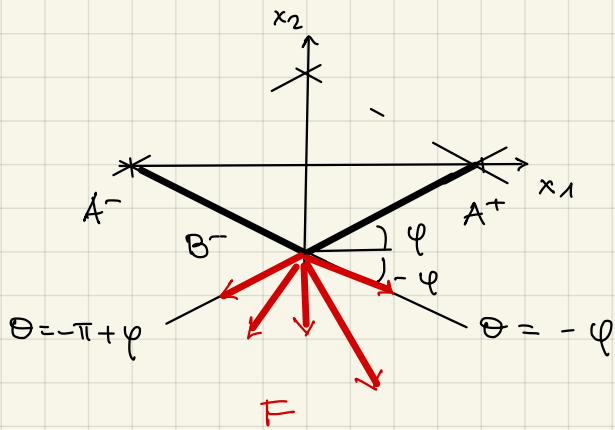


there exists a B^{+} solution

$$\forall F > 0, \forall \Theta \in (\varphi, \pi - \varphi)$$

The case B^- is analogous. $\alpha^- = -\varphi$, $\alpha^+ = \varphi$, $x(l) = -l \sin \varphi e_2$

$$N^- = \frac{F}{2} \left(-\frac{\sin \theta}{\sin \varphi} + \frac{\cos \theta}{\cos \varphi} \right), \quad N^+ = -\frac{F}{2} \left(\frac{\sin \theta}{\sin \varphi} + \frac{\cos \theta}{\cos \varphi} \right)$$

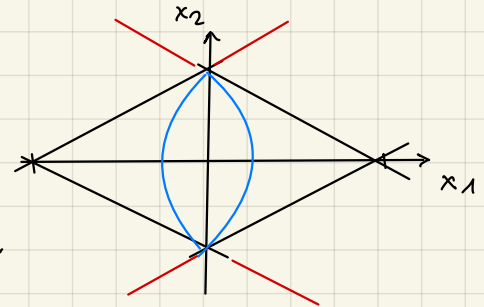


There is a B^- solution

$$\forall F > 0, \theta \in (-\pi + \varphi, -\varphi)$$

We have obtained solutions $\forall F > 0$, $\forall \theta \in (0, 2\pi)$ rework them:

Very nonlinear response, which makes it interesting.



Compare with construction of ellipse, which can be obtained by applying F to a pulley moving along the cable. The pulley spans on (outer) trajectory = ellipse.

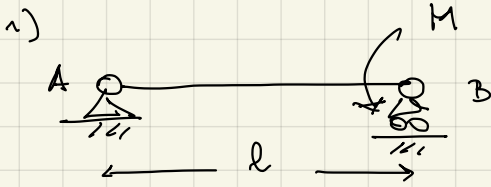
Can reach interior points by varying $2l$ as is the ropes of a sailing boat (scotte)

In closing with whes, say good-bye to this paradigmatic example of deployable structures
from packaging of DNA into spem cells and viral capsids

to catenary and arches of Sagrada Família (Gaudi) and St. Paul's Cathedral (rebuilt by Wren after great fire of London in 1666), to ropes in sail boats to anchor cables several km's long of ocean vessels.

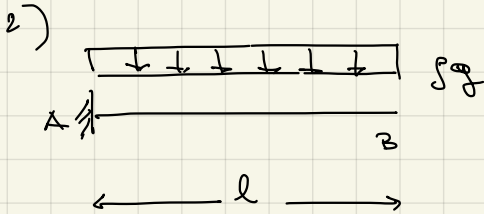
Some exercises on the linear theory

Using linear theory, compute and graph the internal forces, and compute the elastic displacements in the cases sketched below



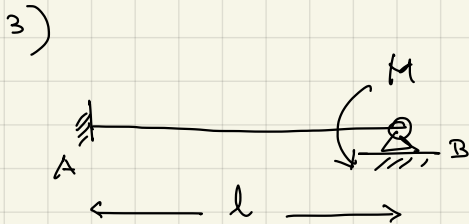
$$\omega_A = ? , \omega_B = ?$$

(rotations at the ends)



$$\omega_B = ? , w_B = ?$$

(rotation and deflection at the end)



$$\omega_B = ?$$

(rotation at end B)